APPLIED PHYSICS
(BASICS OF ELECTRICITY & MAGNETISM)

CHARGING & DISCHARGING:

Charging $\xrightarrow{e^-}$

Discharging $\xleftarrow{e^-}$

INTRODUCTION TO MAGNETISM:

→ Concept given by Greeks (600 BC)

→ In 1820 Oersted gave a concept that electricity can deflect magnetic compass.

→ Micheal Faraday gave concept of Electromagnetism.

→ Maxwell proved it equationally.

Maxwell's Equation: Speed of Light = $c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}$

[ε_0 = permitivity of free space

μ_0 = permeability of free space]

(*)

→ Mechanical waves and Electromagnetic waves.

PROPERTIES OF CHARGE:

→ Same charges repel and opposite charges attract.

→ Charge is quantized.

$q = ne$ [$e^- = 1.6 \times 10^{-19}$ C

$n = \pm 1, \pm 2 \pm 3, \ldots$ (never $0.5e^-$, $1.5e^- \ldots$)]

→ A +ve charge can anhilate -ve charge but a charge cannot simply dissappear.

→ Charge can be continuous along with quantized.

Conservation of charge: $\frac{\partial \rho}{\partial t} + \nabla . J = 0$ [J = current density = I/A]

→ Charge entering = Charge leaving

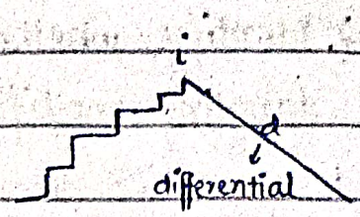
31st Oct, 2024 (LECTURE 2)

APPLIED PHYSICS

ELECTRIC FIELD DUE TO CHARGE DISTRIBUTION (CONTITUOUS)

$q = ne$

Linear charge density = $\lambda = \frac{q}{l}$ = $\sum_{x=1}^{n} i$



Surface charge density = $\sigma = \frac{q}{A}$ — (1)

Volume of charge density = $\rho = \frac{q}{V}$

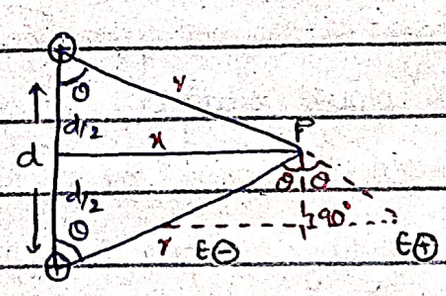
Assignment 1: How can we extract important or precious material from discarded computing devices?

(Due Next Thursday → 8-15 pages)

ELECTRIC FIELD DUE TO DIPOLE:

Electric dipole = $P = qd$

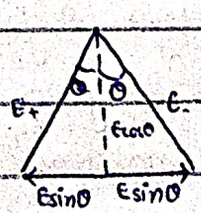
$$E = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{r^2} \quad -(1)$$



Now, horizontal components will cancel each other.

$$E\sin\theta - E\sin\theta = 0$$

So vertical components will support each other.



$$E_{net} = E\cos\theta + E\cos\theta = 2E\cos\theta \quad -(2)$$

By figure: $\cos\theta = \frac{d/2}{r}$

By pythagoras theorem: $r^2 = x^2 + (d/2)^2$

$$r = \left[x^2 + (d/2)^2\right]^{1/2}$$

Using this in (2): $E = 2 \times \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{\left[x^2 + (d/2)^2\right]} \cdot \frac{d/2}{\left[x^2 + (d/2)^2\right]^{1/2}}$

$$= \frac{1}{4\pi\varepsilon_0} \cdot \frac{qd}{[x^2 + (d/2)^2]^{3/2}}$$

$$= \frac{P}{4\pi\varepsilon_0} \cdot \frac{1}{x^3 (1 + (d/2x)^2)^{3/2}}$$

$$= \frac{P}{4\pi\varepsilon_0} \times \frac{[1 + (d/2x)^2]^{-3/2}}{x^3}$$

Consider: $\left[1 + \left(\frac{d}{2x}\right)^2\right]^{-3/2}$ $\quad (1+x)$ $\quad \left[(1+x)^n = 1 + nx + \frac{n(n-1)x^2}{2!} + \dots\right]$

$$= 1 + \left(\frac{-3}{2}\right)\left(\frac{d}{2x}\right)^2 + \dots \quad (\text{neglecting higher terms})$$

($\left(\frac{d}{2x}\right)^2 \to 0$)

$$\boxed{E = \frac{1}{4\pi\varepsilon_0} \cdot \frac{P}{x^3}}$$

ELECTRIC FIELD DUE TO RING OF CHARGE:

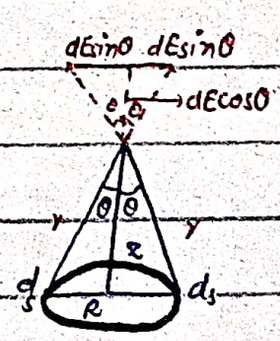
$$E = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{r^2}$$

$$= \frac{1}{4\pi\varepsilon_0} \times \frac{\lambda ds}{(z^2 + R^2)} \quad -①$$

$$\lambda = \frac{q}{l}$$

$$q = \lambda l$$

$$(l = d_s)$$



d_s = small patch

$$dE_z = dE\cos\theta \quad -②$$

$$E_z = \int dE\cos\theta$$

$$\cos\theta = \frac{z}{r}$$

$$E_z = \int \frac{1}{4\pi\varepsilon_0} \times \frac{\lambda ds}{(z^2 + R^2)} \times \frac{z}{(z^2 + R^2)^{1/2}} \qquad \cos\theta = \frac{z}{(z^2 + R^2)^{1/2}}$$

$$= \frac{\lambda z}{4\pi\varepsilon_0 (z^2 + R^2)^{3}} \int ds$$

$$= \frac{\lambda z (2\pi R)}{4\pi\varepsilon_0 (z^2 + R^2)^{3/2}} \qquad \int ds = 2\pi R$$

$$E_z = \frac{zq}{4\pi\varepsilon_0 (z^2 + R^2)^{3/2}} \qquad [\lambda = q/l \Rightarrow q = \lambda l \Rightarrow q = \lambda 2\pi r]$$

$= \frac{2q}{4\pi\varepsilon_0 \times x^3}$ ($x >> L$ so neglecting it)

$$E_x = \frac{q}{4\pi\varepsilon_0 x^2}$$

Nov 2, 2024

PHY : APPLIED PHYSICS

ELECTRIC FIELD ON DISK OF CHARGE:

$$E = \frac{1}{4\pi\varepsilon_0} \cdot \frac{\lambda ds}{r^2}$$

$$dE_z = \frac{1}{4\pi\varepsilon_0} \times \frac{\sigma(2\pi w)dw}{r^2}$$

$$= \frac{1}{4\pi\varepsilon_0} \times \frac{\sigma(2\pi w)dw}{(w^2+z^2)^{3/2}}$$

Integrating: $$E = \frac{\sigma z}{4\pi\varepsilon_0} \int (w^2+z^2)^{-3/2} (2w)\, dw$$

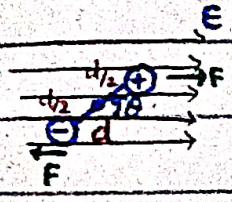
(Let $x = w^2+z^2$ and $dx = 2w\,dw$)

$$E = \frac{\sigma z}{4\varepsilon_0} \left| \frac{-2}{(w^2+z^2)^{1/2}} \right|_0^R$$

$$= \frac{\sigma z}{4\varepsilon_0} \left| \frac{-2}{(R^2+z^2)^{1/2}} + \frac{2}{(z^2)^{1/2}} \right|$$

$$= \frac{\sigma z}{2\varepsilon_0} \left| \frac{1}{z} - \frac{1}{(R^2+z^2)^{1/2}} \right|$$

Dipole in an Electric Field:



$$\tau = \tau_1 + \tau_2$$

$$= d/2\, F\sin\theta + d/2\, F\sin\theta$$

$$= dF\sin\theta$$

$$= d(qE)\sin\theta \qquad (F = qE)$$

$$\tau = (dq)E\sin\theta \quad \Rightarrow \quad (P = qd)$$

$$\tau = PE\sin\theta$$

$$\vec{\tau} = \vec{P} \times \vec{E}$$

$s = r\theta$ (arc length)

(small path)

work done: $$W = \int_{\theta_i}^{\theta_f} d\omega$$ (Due to opposite direction rotation (clockwise) or it tends to decrease)

$$W = -\int_{\theta_i}^{\theta_f} \tau \, d\theta \qquad \text{(Linear to Rotatory)}$$

$$= -\int_{\theta_i}^{\theta_f} PE \sin\theta \, d\theta$$

$$= |PE\cos\theta|_{\theta_i}^{\theta_f}$$

$$= PE\cos\theta_f - PE\cos\theta_i$$

(Let $\theta_i = 90^\circ$)

$$W = PE\cos\theta_f$$

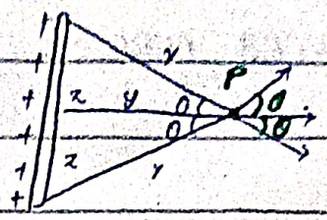
$$= PE\cos\theta$$

November 7, 2024 (Lecture 4)

APPLIED PHYSICS

ELECTRIC FIELD ON AN INFINITE LINE OF CHARGE:

$$dE = \frac{1}{4\pi\varepsilon_0} \cdot \frac{\lambda dz}{r^2}$$



$$dE = \frac{1}{4\pi\varepsilon_0} \cdot \frac{\lambda dz}{(y^2+z^2)}$$

$$dE_z = dE\cos\theta \quad -(a)$$

Integrating eq. (a): $E_z = \int dE_z = \int dE\cos\theta$

$$E_z = \frac{1}{4\pi\varepsilon_0} \int_{-\infty}^{+\infty} \frac{\lambda dz}{(y^2+z^2)} \cdot \cos\theta$$

$$= \frac{1}{4\pi\varepsilon_0} \times 2 \int_0^{\infty} \frac{\lambda dz}{y^2+z^2} \cdot \cos\theta \quad -(b)$$

Let $z = y\tan\theta$ $\Rightarrow$ $z^2 = y^2\tan^2\theta$

$dz = y\sec^2\theta\, d\theta$

(Using Substitution method)

Eq. (b): $E_z = \frac{\lambda}{2\varepsilon_0\pi} \int_0^{\pi/2} \frac{y\sec^2\theta}{y^2+y^2\tan^2\theta} \cdot \cos\theta\, d\theta$

$\left(\because \tan\frac{\pi}{2} = \infty \text{ so replaced it}\right)$

$$= \frac{\lambda}{2\pi\varepsilon_0} \int_0^{\pi/2} \frac{y\sec^2\theta}{y^2(1+\tan^2\theta)} \cdot \cos\theta\, d\theta$$

$$= \frac{\lambda}{2\pi\varepsilon_0} \int_0^{\pi/2} \frac{y\sec^2\theta}{y(\sec^2\theta)} \cdot \cos\theta\, d\theta$$

$$= \frac{\lambda}{2\pi\varepsilon_0 y} \int_0^{\pi/2} \cos\theta\, d\theta$$

$$E_x = \frac{\lambda}{2\pi\varepsilon_0 y} |\sin\theta|_0^{\pi/2}$$

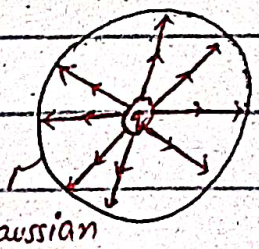
$$\boxed{E_x = \frac{\lambda}{2\pi\varepsilon_0 y}}$$

GAUSS'S LAW:

German mathematician → : Fredrick Gauss

BASIC DEFINITION:

Total charge enclosed by the surface divided by ε_0 is equal to electric flux.



$$\Phi_E = \frac{q_{enc}}{\varepsilon_0} \quad -(a)$$

$$\Phi_E = \oint E \cdot dA \quad -(b)$$

Comparing (a) & (b):

Deriving Coulomb's Law using Gauss' Law:

$$\frac{q_{enc}}{\varepsilon_0} = \oint E \cdot dA$$

$$\frac{q_{enc}}{\varepsilon_0} = E(4\pi r^2)$$

$$E = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q_{enc}}{r^2}$$

(As $E = F/q \Rightarrow F = q_0 E$)

$$F = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q_{enc} \cdot q_0}{r^2} \quad \text{(Coulomb Law)}$$

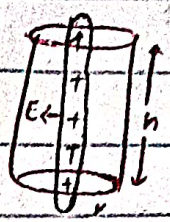
APPLICATIONS OF GAUSS LAW:

→ INFINITE SHEET of CHARGE:

→ INFINITE LINE OF CHARGE:

INFINITE LINE OF CHARGE:

$$\frac{q_{enc}}{\varepsilon_0} = \oint E \cdot dA$$

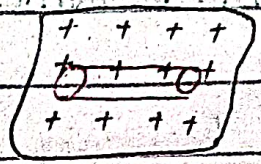


$$\frac{q_{enc}}{\varepsilon_0} = E(2\pi r h)$$

$$\frac{\lambda h}{\varepsilon_0} = E(2\pi r h) \Rightarrow \boxed{E = \frac{\lambda}{2\pi r \varepsilon_0}}$$

INFINITE SHEET OF CHARGE:

$$\frac{q_{enc}}{\varepsilon_0} = \oint E \cdot dA$$



$$\frac{\sigma A}{\varepsilon_0} = (EA + EA)$$

$$\sigma A = \varepsilon_0 (2EA) \Rightarrow \boxed{E = \frac{\sigma}{2\varepsilon_0}}$$

APPLIED PHYSICS

1) The average distance (r) between the electron and proton in the H-atom is 5.3×10^{-11} m. (a) What is the mag. of avg. Electrostatic force that is acts between these two particles? (b) what is the mag. of avg. Gravitational force that acts between these two particles?

2) An electron placed near a charged body experiences a force in the +ve y direction of Magnitude 3.6×10^{-8} N.

(a) What is the electric field at that location?

(b) What is the value of $\vec{F}$ exerted by the same charged body on the α-particle ($q = +2e$) placed at the location formally occupied by electron?

3) The magnitude of the average $\vec{E}$ present in the Earth's atmosphere just above the surface of Earth is about 150 NC^{-1} directed towards the centre of the Earth. What is the total net surface charge carried by Earth? (Assume Earth to be a Conductor)

SOLUTION:

(1) **DATA:**

Distance = $r = 5.3 \times 10^{-11}$ m

Charge on proton = $p^+ = 1.6 \times 10^{-19}$ C

Charge on electron = $e^- = -1.6 \times 10^{-19}$ C

Mass of proton = $m_p = 1.6727 \times 10$

Mass of electron = $m_e = 9.1 \times 10^{-31}$ kg

Electrostatic Constant = $k = 9 \times 10^9$ Nm^2C^{-2}

Gravitational Constant = $G = 6.673 \times 10^{-11}$ Nm^2kg^{-2}

TO FIND:

(a) Electrostatic force = F_E = ?

(b) Gravitational force = F_G = ?

SOL:

(a) $F_e = \frac{kq_1q_2}{r^2}$

$= \frac{(9\times10^9)(1.6\times10^{-19})^2}{(5.3\times10^{-11})^2}$

$= 8.2\times10^{-8}\ N$

(b) $F_g = \frac{G m_1 m_2}{r^2}$

$= \frac{(6.673\times10^{-11})(1.672\times10^{-27})(9.1\times10^{-31})}{(5.3\times10^{-11})^2}$

$= 3.45\times10^{-47}\ N$

(2) DATA: Electric force = $F_e = 3.6\times10^{-8}\ N$

Charge on electron = $q_e = 1.6\times10^{-19}\ C$

Charge on α-particle = $q_\alpha = 2(1.6\times10^{-19})$

$= 3.2\times10^{-19}\ C$

TO FIND: (a) Electric field = $E = ?$

(b) Electric force = $F_e = ?$

(a) $E = \frac{F}{q} = \frac{3.6\times10^{-8}}{1.6\times10^{-19}} = 2.25\times10^{11}\ NC^{-1}$

(b) $F = q_\alpha E$

$F = (3.2\times10^{-19})(2.25\times10^{11})$

$F = 7.2\times10^{-8}\ N$

(3) DATA: Magnitude of Electric Field = $E = 150\ NC^{-1}$

$\varepsilon_0 = 8.84\times10^{-12}\ J$

Radius = $r = 6.4\times10^6\ m$

TO FIND:

Charge = $q = ?$

SOL:

$\Phi = EA = E(4\pi r^2)$ ... (i)

$\Phi = \frac{q}{\varepsilon_0}$ ... (ii)

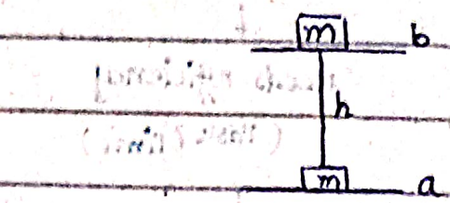
$$\frac{q}{\varepsilon_0} = E(4\pi r^2)$$

$$\frac{q}{\varepsilon_0} = (150)(4\pi)(6.4\times10^{6})^2$$

$$q = (150)(4\pi)(6.4\times10^{6})^2(8.84\times10^{-12})$$

$$q = 6.823\times10^{5}\ C$$

ELECTRIC POTENTIAL ENERGY: (U)

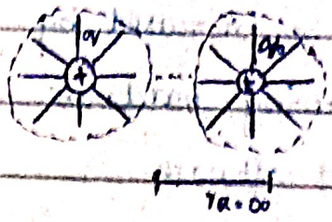


$$W = F.d$$

$$U = F.d = -mgh$$

$$U_b - U_a = -\int_a^b F.ds$$

$(F = qE)$



$$U_b - U_a = -q\int_{r_a}^{r_b} E.dr$$

$$= -q\int_{r_a}^{r_b} \frac{1}{4\pi\varepsilon_0}\cdot\frac{q_0}{r^2}\,dr$$

$$= \frac{-qq_0}{4\pi\varepsilon_0}\int_{r_a}^{r_b}\frac{1}{r^2}\,dr$$

$$= \frac{-qq_0}{4\pi\varepsilon_0}\left[\frac{-1}{r}\right]_{r_a}^{r_b}$$

$$= \frac{qq_0}{4\pi\varepsilon_0}\left|\frac{1}{r_b}-\frac{1}{r_a}\right|$$

$$= \frac{qq_0}{4\pi\varepsilon_0}\times\frac{1}{r_b} \qquad (r_a=\infty,\ 1/r_a=0)$$

$$U_r = \frac{1}{4\pi\varepsilon_0}\times\frac{qq_0}{r_b}$$

$$\int\frac{1}{r^2}\,dr = \int r^{-2}\,dr$$

$$= \frac{r^{-1}}{-1} = -\frac{1}{r}$$

28th Nov, 2024 (Lec 6)

APPLIED PHYSICS

ELECTRIC POTENTIAL:

Potential Energy per unit charge.

$$V_b - V_a = \frac{U_b - U_a}{q_0}$$

$$\Delta V = \frac{\Delta U}{q_0} \qquad \text{(unit is } JC^{-1}\text{)}.$$

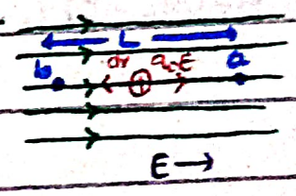
$$V = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{r}$$

CALCULATING THE POTENTIAL FROM THE FIELD:

$$W_{ab} = \int F\,dx$$

$$= \int (-q_0 E)\,dx$$

$$W_{ab} = -q_0 E L \quad -(a)$$



$\vec{E}$ = External Electric field

We know:

$$V_b - V_a = \frac{U_b - U_a}{q_0} \quad -(b)$$

$$U_b - U_a = -W_{ab}$$

So (b) becomes:

$$V_b - V_a = -\frac{W_{ab}}{q_0}$$

$$V_b - V_a = \frac{q_0 E_0 L}{q_0} \qquad \text{(From a)}$$

$$V_b - V_a = EL$$

$$V_b - V_a = E \cdot L \qquad \int$$

$$V_b - V_a = -\int E \cdot dL$$